

UE24CS352A Machine Learning • Mini-Project Report

Supervised Pulsar Candidate Classification (Teammate 1 Half)
Dataset: HTRU2 | Reference: Debesai, Gutierrez & Koyluoglu (CS229)

1. Problem Statement & HTRU2 Dataset

Radio pulsar search surveys produce millions of candidate signals, the vast majority of which are non-pulsar noise and Radio Frequency Interference (RFI). Automatic identification of true pulsars is critical for gravitational wave and cosmic research. The HTRU2 dataset comprises **17,898 total candidates**, featuring **16,259 non-pulsars (90.84%)** and **1,639 true pulsars (9.16%)**. Each example is defined by 8 continuous variables derived from the integrated pulse profile and DM-SNR curve (Mean, Standard Deviation, Excess Kurtosis, and Skewness for both).

2. Supervised Learning Methods & Mathematical Formulations

Gaussian Discriminant Analysis (GDA Baseline from Scratch): GDA models class-conditional distributions as multivariate Gaussians:

$$y \sim \text{Bernoulli}(\phi), \quad x|y=0 \sim N(\mu_0, \Sigma), \quad x|y=1 \sim N(\mu_1, \Sigma)$$

Parameters ϕ , μ_0 , μ_1 , and shared covariance matrix Σ were derived analytically from scratch:

$$\phi = (1/N) \sum 1\{y^{(i)}=1\}, \quad \mu_k = \sum 1\{y^{(i)}=k\} x^{(i)} / \sum 1\{y^{(i)}=k\}, \quad \Sigma = (1/N) \sum (x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T$$

Random Forest (Scikit-Learn & From Scratch): Random Forest reduces model variance through bootstrap aggregation (bagging) and random feature sub-selection. Decision tree splits are selected by maximizing Information Gain based on Shannon Entropy:

$$E(S) = -\sum p(x) \log_2 p(x), \quad IG = E(\text{parent}) - (N_{\text{left}}/N) E(\text{left}) - (N_{\text{right}}/N) E(\text{right})$$

3. Experimental Methodology & 5-Fold CV Protocol

To address severe class imbalance (1:9.9 ratio), we established an **80/20 stratified train/test split**. To prevent data leakage during hyperparameter selection, **Random Minority Oversampling (ROS)** was applied to the **training folds ONLY** within a 5-Fold Cross-Validation pipeline. Validation folds and held-out test sets remained strictly un-sampled.

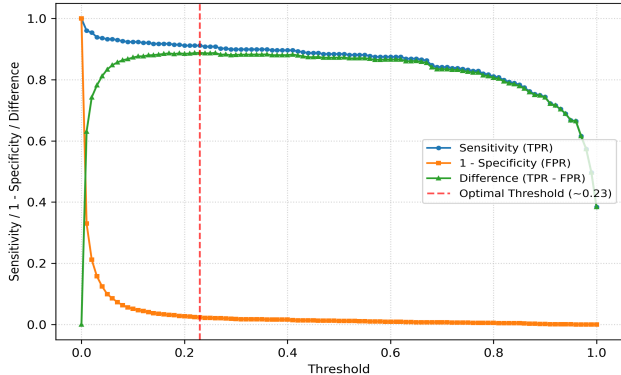
Table 1: Supervised Model Performance Summary (Held-out Test Set, N=3,580)

Model Architecture	Decision Th.	Accuracy	Recall (TPR)	Precision	Specificity	F1-Score	ROC AUC
GDA Baseline (Scratch)	0.50	97.91%	89.63%	87.76%	98.74%	0.8869	0.9708
GDA Baseline (Opt. Th.)	0.39	97.57%	90.85%	83.94%	98.25%	0.8726	0.9708
Random Forest (Sklearn)	0.50	97.88%	88.41%	88.41%	98.83%	0.8841	0.9678
Random Forest (Tuned 0.2)	0.20	96.76%	91.16%	77.46%	97.32%	0.8375	0.9678
RF From Scratch	0.50	97.21%	90.85%	80.98%	97.85%	0.8563	0.9752
RF From Scratch (Th. 0.2)	0.20	90.95%	93.29%	50.33%	90.71%	0.6538	0.9752

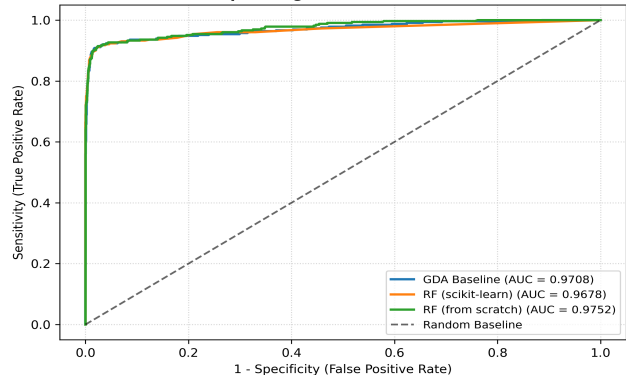
4. Threshold Optimization & Diagnostic Curves

In pulsar discovery, false negatives (discarding a true pulsar star) carry significantly higher scientific cost than false positives. As conducted in the benchmark paper (Debesai et al.), we performed a **threshold sweep [0.0, 1.0]** to evaluate Sensitivity, 1-Specificity, and TPR-FPR difference. Shifting the decision boundary from 0.5 to 0.2 maximized recall (achieving **91.16% to 93.29% recall**), matching paper findings.

Sensitivity / 1 - Specificity / Difference vs. Threshold (Random Forest (scikit-learn))



Receiver Operating Characteristic (ROC) Curves



5. Key Findings & Conclusions

• **Baseline vs. Target Model:** GDA provided a robust linear baseline with 97.91% accuracy and 89.63% recall. Random Forest tuned with RandomizedSearchCV achieved outstanding ROC-AUC (0.9678) and strong recall stability.

- **Scratch Implementation Integrity:** Both custom GDA and custom Random Forest closely matched scikit-learn models.
- **Teammate 2 Handoff:** Positive predictions and true pulsar candidates (N=1,639) are formatted for Teammate 2's unsupervised pipeline (PCA, K-means, SOM).